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Exploratory/Confirmatory Factor Analysis and Scale Development

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A considerable portion of management research uses employee surveys in which respondents are asked to describe their experiences at their workplace via questionnaires. These surveys can be used to test theories, assess workplace climate, or evaluate interventions or changes in organizational policies. While theory describes the relations among constructs examined, statistical analysis of the employee-provided data is ultimately used to reach conclusions about construct relations. A key to the success of the measurement process is that the items used to represent the construct are, in fact, good representations of the constructs, which is to say they have construct validity. Within this context, exploratory and confirmatory factor analyses play an important role, as both techniques yield statistical information that can be used to judge the quality of the resulting items and their value as measures of their presumed underlying constructs.

Success at improving construct validity in organizational research has been facilitated by methodologists who focus on what happens before the items of the questionnaire are presented to the sample whose responses will be included in the factor analysis used to develop and refine the scales to be used. For example, Podsakoff, MacKenzie, and Podsakoff (2016) have emphasized the importance of having clear conceptual definitions of constructs. Next in the research process is the development of specific items, where the judgment needs to be made that the content of the items matches the definition of the construct and, if so, the researcher concludes they have content validity (see Colquitt, Sabey, Rodell, & Hill, 2019).

It has long been recognized that there is value in combining items to form scales, even when concepts are clearly defined and items are demonstrated to have content validity. When multiple constructs are included in the research design, evidence is considered as to whether items that are intended to measure a common construct are highly correlated with each other. It is also desired that these same items not correlate highly with items of different constructs. Exploratory and confirmatory factor analyses (EFA, CFA) are data analysis tools for evaluating a matrix of correlations based on items measuring multiple constructs. Relevant evidence from such a factor analysis can inform decisions about (a) items that are intended to measure the same thing, (b) if what they are measuring is something different from other things being measured, and (c) ultimately, how many different things are being measured.

Increased use of EFA and CFA in organizational research increases the importance of key decision points in their implementation. Statistical

developments with both methods have created more choices for researchers using one or both techniques, and researchers have new software programs available to use. This leads to a challenge for those wanting to follow best practices as they evaluate items and develop scales for use in their theory-testing or workplace environment assessment. The purpose of this chapter is to provide an overview of current EFA/CFA methods as used for scale development, including an organizational-based example that demonstrates these practices with readily available software. Our presentation of this example will highlight key differences and similarities between the models underlying EFA and CFA while advocating that both can, should, and typically are used in a confirmatory way.

EFA/CFA EXAMPLE: TRUSTWORTHINESS

Preliminary Analysis

In the following example, we use data collected from Hanna (2021) that measured the construct of trustworthiness. Extant research conceptualizes trustworthiness as a multidimensional construct consisting of three facets – ability (Abil), benevolence (Bene), and integrity (Inte), (Frazier, Johnson, Gavin, Gooty, & Snow, 2010). The trustworthiness measure we used includes nine items – three items reflecting each of the three dimensions – as utilized by Jones and Shah (2015). Our labels for the items will be Ab1-Ab3, Be1-Be3, and In1-In3. Our analyses for this example used the "lavaan," "psych," "GPArotation," and "corrplot" R packages. These packages must be installed and activated to run the included syntax. We used R-based packages for our example because they are freely available, are increasingly popular, and can be used to conduct preliminary data management and evaluation as well as both exploratory and confirmatory factor analyses within the same general platform. We also checked the necessary properties of the data (for a comprehensive list, see Watkins, 2018).

We next conducted a subjective examination of the correlations among items to ensure that conducting an EFA/CFA is appropriate. As noted in our introduction, since with our analysis we hope to find support for three dimensions, we expect that the three items of ability will be highly correlated with each other, as will the correlations among the three benevolence items and the three integrity items. We also anticipate that the correlations between items across dimensions will be lower. This pattern of three distinct clusters of correlations is needed for the three factors to emerge and be supported in both the EFA and CFA. An examination of these correlations prior to the EFA/CFA can provide preliminary support for our proposed underlying dimensionality. More importantly, if the desired pattern of correlations is not obtained, these unexpected values can help us understand problems that emerge in the subsequent multivariate analysis.

If beginning with a dataset that contains items not used in the EFA, the first step is to create a data frame with only the items to be included. To produce this correlation matrix, begin with the following syntax:

```
[1] names(Data_Full)
[2] TWData <- Data_Full[,c(78,79,80,84,85,86,90,91,92)
[3] TWCorMat <- cor(TWData, use="pairwise.
complete.obs")
```

Using names(Data_Full) will produce a list of all items contained in the full dataset and a column number assigning its place in the dataset. In our example, the full dataset is named "Data_Full." Once the correct items and their corresponding column numbers are identified, a new data frame containing only those items can be created. To do this, choose a name to assign the data frame – since we are examining trustworthiness, our data frame is "TWData" – and use the syntax on line 2. This uses R's concatenate function to create the data frame using only the columns specified from the original dataset – in our case, columns 78-80, 84-86, and 90-92. The syntax on line 3 will create a correlation matrix object in our R environment – "TWCorMat" – for later use. The data frame created in line 2 can then be used to create a correlation matrix among items using the following:

```
[4] TWCor.rcorr <- rcorr(as.matrix(TWData))
[5] TWCor.rcorr
```

This syntax will create a new object in the R environment – in this case, "TWCor.rcorr" – representing the correlation matrix produced by line 1. Line 5 produced a correlation matrix among the nine items shown in Table 18.1.

A subjective examination of the strength of correlations among items suggests preliminary support for the presence of the proposed dimensions of ability, benevolence, and integrity. The three sets of within-dimension correlations are relatively large (.61, .71, .65; .72, .41, .50; .45, .48, .49) and are of greater value than the 27 between-dimension correlations. Another useful approach is to visualize the item correlations with R's "corrplot" package, using the following syntax:

Table 18.1 Correlation matrix for trustworthiness items

	Ab1	Ab2	Ab3	Be1	Be2	Be3	In1	In2	In3
Ab1	1.00	0.61	0.71	0.16	0.13	0.21	0.26	0.24	0.31
Ab2	0.61	1.00	0.65	0.27	0.26	0.25	0.25	0.27	0.26
Ab3	0.71	0.65	1.00	0.24	0.18	0.27	0.30	0.28	0.27
Be1	0.16	0.27	0.24	1.00	0.72	0.41	0.32	0.32	0.26
Be2	0.13	0.26	0.18	0.72	1.00	0.50	0.38	0.36	0.30
Be3	0.21	0.25	0.27	0.41	0.50	1.00	0.52	0.34	0.40
In1	0.26	0.25	0.30	0.32	0.38	0.52	1.00	0.45	0.48
In2	0.24	0.27	0.28	0.32	0.36	0.34	0.45	1.00	0.49
In3	0.31	0.26	0.27	0.26	0.30	0.40	0.48	0.49	1.00

[6] corrplot(TWCorMat)

This syntax uses the correlation matrix created earlier to visualize the correlations among items in the form of a correlogram (see Appendix 18.A). The magnitude of the correlations is represented via the shade of the items' intersection.

EXPLORATORY FACTOR ANALYSIS: BACKGROUND

Since the desired patterns are obtained in the correlations, we continued with our analyses and conducted the EFA. A path diagram associated with our three-factor model of trustworthiness is presented in Figure 18.1. Although path diagrams for EFA models are not typically used, we present ours to illustrate key differences between EFA and CFA models, the latter of which are often accompanied by path diagrams. Following tradition, the three circles represent the factors of ability, benevolence, and integrity, labeled as T-Ab, T-Be, and T-In. For an EFA, these factors are assumed to be standardized with variance, this is shown in the diagram using arrows with *1.0, which leads to estimates of the relations among the three factors to be standardized and referred to as factor correlations (double-headed curved arrows between factors). Also shown in the figure is the assumption that each of the three factors is presumed to influence all nine of the items (Ab1-In3). This influence is shown via the factor loadings (single-headed arrow from factor to item). Finally, the path diagram shows that in addition to the influence of the three factors, each item has unique variance that is not accounted for by the factors (u_1-u_9).

Given the above, the equation for the first item can be shown to be:

[7] $Ab1 = FL\ (T\text{-}Ab) + FL\ (T\text{-}Be) + FL\ (T\text{-}In) + u_i$

The factor loadings of the equation (FL) can be interpreted as standardized regression coefficients, in which the item is the DV and the factors are the IVs. Variance in Ab1 due to Ab, Be, and In is referred to as common variance and u_i is unique variance, consisting of random and systematic errors. Similar equations for the remaining eight items are also implemented in the analysis. Our goals of the EFA include to determine (a) if there is support for the proposed three factors, (b) how much distinct common variance in each item is associated with each factor, (c) how strongly each item is linked to each factor, and (d) how highly correlated are the three factors. We will examine the types of evidence linked to each of these goals.

The factor analysis begins with the already described correlation matrix R among the nine items Ab1- In3. As a first step, the diagonal of R is "adjusted," in that the 1.0 values are replaced with Squared Multiple Correlations (SMC) for each item, in which the item is used as the dependent variable and all other items are used as independent variables. It is this step that distinguishes EFA (sometimes referred to as common factor analysis) from principal component analysis, which assumes no item unique variance and is implemented using the original correlation matrix. These SMC values are referred to as "initial" communalities, they reflect the presence of u_i and are an estimate of the common variance in each item due to the three factors, and as mentioned the adjustment is what distinguishes EFA from principal components analysis.

Next, the analysis subjects the resulting adjusted correlation matrix to advanced matrix algebra as R^A is "diagonalized." Simply, this process leads to an expression of the shared variance in the correlations among the items in a different way. This process involves additional matrices of

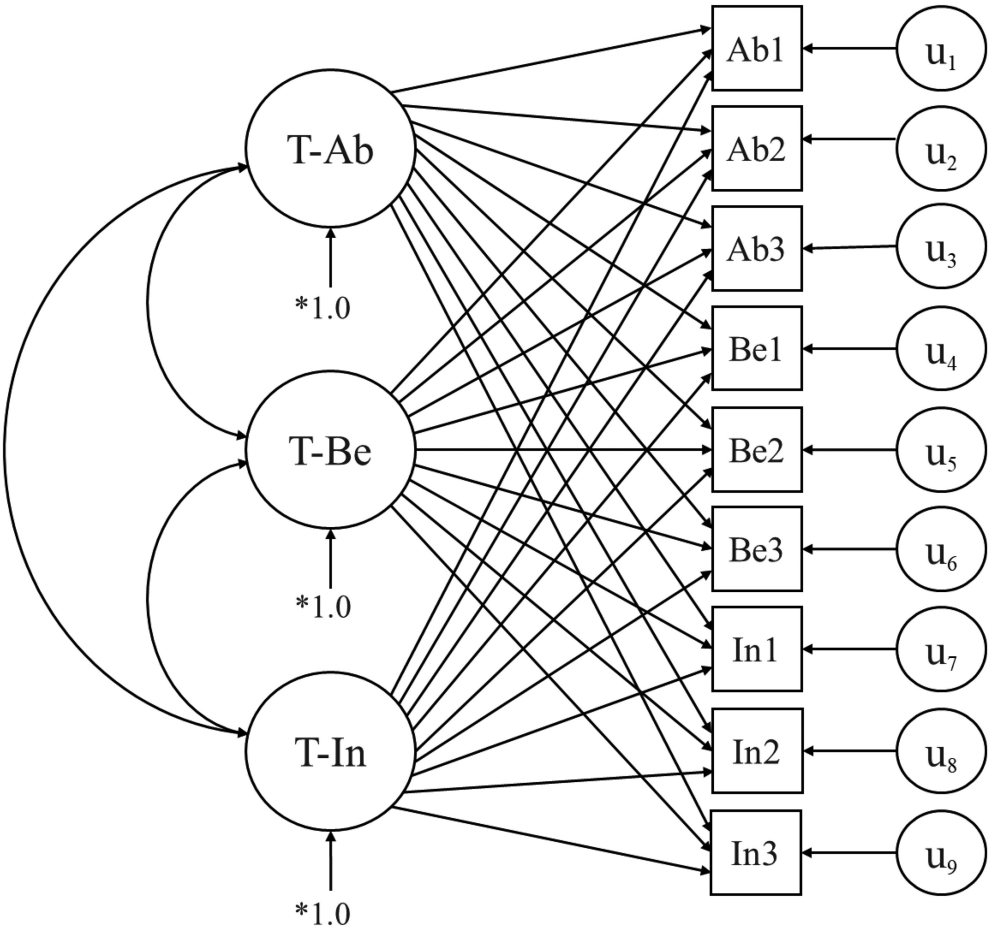


Figure 18.1 Path diagram EFA model with trustworthiness items

eigenvalues and eigenvectors, and their relationship to R^A can be expressed as:

[8] $EVect \times R^A \times EVect^T = EVal$

Eval is a diagonal matrix of eigenvalues (all off-diagonal elements are zero) and each reflects decreasing amounts of shared variance among the items in sets. These eigenvalues are obtained sequentially, and with each one, the amount of shared variance decreases. While the total number of eigenvalues will equal the number of items, in our case nine, our analysis focuses on the first three and their associated eigenvectors since our model proposed three dimensions. Our interest will be in how many eigenvalues are greater than 1.0, how many will lie above a line through the lower eigenvalues referred to as the scree plot, and how they obtained eigenvalues from the EFA compared to those that would be obtained from a similar size correlation matrix (through the use of parallel analysis). In our case, we will hope that these three pieces of information support the claim that three factors are present in the nine items.

As shown earlier for each eigenvalue there is an associated eigenvector, and each of these is subjected to more matrix algebra that results in a set of "unrotated" factor loadings as shown in line [7] for Ab1 presented above. The unrotated factor loadings (FL) are obtained as:

[9] $FL = EVect^*(EVal)^{1/2}$

Another transformation referred to as rotation is implemented (linked to the geometric interpretation

of factor analysis), which magnifies the differences between the "rotated" factor loadings (larger factor loading values increase, smaller ones decrease). For the process of rotation, the researcher must choose between assuming the factors are either uncorrelated (orthogonal) or correlated (oblique). Researchers often believe that the assumption of correlated factors is theoretically justified, as we did with our trustworthiness example. These factor loadings from an oblique rotation are often called "pattern coefficients," and since the factors and the items have unit variances they can be interpreted as standardized regression coefficients. If our proposed model is correct, a pattern for each equation's factor loadings will occur referred to as "simple structure," which can be described as for each item, one factor loading will be large and the other two will be small.

With the above background in mind, we conducted our EFA, and given our theoretical model, we expected to find support for three factors. This support would include the desired pattern of eigenvalues, including three values greater than 1.0 and three above a "scree" line through all the eigenvalues. Also relevant is a comparison of the obtained eigenvalues based on those suggested by the parallel analysis, simple structure among the factor loadings, and factor correlations that are not too high to challenge the discriminant validity of the three factors. We emphasize the importance of being dispassionate and objective in reviewing factor analysis results, while trying to avoid "confirmatory bias" in the interpretation of the results. We also note that given the challenges in making such a complex judgment, it can also be useful to compare one's proposed model with alternative models with one less and one more factor, hoping the latter two will be found less desirable.

EXPLORATORY FACTOR ANALYSIS: RESULTS WITH TRUSTWORTHINESS

The following syntax was used to conduct an EFA:

```
[10] TWEFA <- fa(r=TWCor.rcorr, nfactors=3,
rotate="oblimin," fm="ml")
```

When writing this syntax, we can see that several decisions must be made by the researcher, keeping a few previous points in mind. First, remember that theory should inform the number of proposed dimensions, represented in the syntax as nfactors=3. Conceptually, trustworthiness includes three dimensions, influencing our decision to set the number of factors to three. Next, we chose the appropriate factor rotation, specified in the syntax as rotate="oblimin." Oblimin rotation is used when factors are assumed to be correlated, which is appropriate given the three factors are dimensions of the overarching trustworthiness construct. Finally, the factoring method must be specified, represented in our syntax as fm="ml." We suggest using maximum likelihood factor analysis due to its efficiency and flexibility in situations where other methods' assumptions may be violated (White, 1982). We also included code to produce the eigenvalues, which as noted earlier provide important information to help evaluate the proposed model. We used the following syntax:

```
[11] TWEigen <- eigen(TWCorMat)
[12] print(TWEigen)
```

The syntax on line 12 created an object – in this case, "TWEigen" – that used the correlation matrix object created at the beginning of the project to produce the eigenvalues for all items included in the correlation matrix. Note that the number of eigenvalues should always be equal to the number of items in the data frame. Running the syntax included above led to output that consists of several essential pieces of information as results, and these are shown in Table 18.2.

As we forced the extraction of three factors based on our theory, the first three columns shown above are the factor loadings of each item onto each of the three factors. As noted earlier, the magnitude of these factor loadings provides a first look at whether the items follow the proposed dimensionality informed by our theory. The desired pattern of simple structure emerged, as shown by the highlighted clusters in each of the three columns. We found that each item had a high factor loading in one column and noticeably lower loadings in the others, highlighting the discriminant nature of each. Also, it is important that each of the three sets of items had high loadings on the factor defined by the item content – for example, the three ability items each had high loadings on the same factor.

Other important information in the output included the columns marked "h2" and "u2." The first column (h2) provides information about the amount of variance in each item explained by the included factors, the item's communality. The second column (u2) is the residual variance, the item's uniqueness. It is important to note that sum of each item's communality and uniqueness should always equal one (i.e., u2 = 1 – h2). Also, this output shows the correlations among factors,

Table 18.2 Results of EFA for trustworthiness items

Eigen values:								
3.8085	1.6342	1.0292	0.6612	0.4990	0.4435	0.3812	0.2976	0.2457
Standardized loadings (pattern matrix):								
	Abil	Bene	Inte	h2	u2			
Ab1	0.82	0.04	–0.08	0.68	0.32			
Ab2	0.74	–0.06	0.14	0.58	0.42			
Ab3	0.86	0.02	–0.01	0.74	0.26			
Be1	0.08	0.00	0.75	0.59	0.41			
Be2	–0.03	0.02	0.93	0.88	0.12			
Be3	0.00	0.27	0.48	0.44	0.56			
In1	–0.01	0.72	0.02	0.54	0.46			
In2	0.03	0.58	0.06	0.40	0.60			
In3	0.03	0.73	–0.08	0.49	0.51			
Factor correlations:								
	Abil	Bene	Inte					
Abil	1.00	0.46	0.25					
Bene	0.46	1.00	0.53					
Inte	0.25	0.53	1.00					

shown in the final 3×3 matrix. It is essential to note the strength of the correlations among factors, as factor correlations that are too high might suggest problems with discriminant validity. As previously noted, when reviewing these values, the number of eigenvalues that exceed a magnitude of 1.0 should be noted, as this number informs the number of supposed factors to the construct (Marsh et al., 2009). In our example, three eigenvalues exceed 1.0, suggesting that the trustworthiness construct contains three distinct dimensions. A scree plot of the successive eigenvalues previously generated was also examined to avoid limitations of the eigenvalue 1.0 cut-off approach. A scree plot was produced with the following syntax:

```
[13] TWScree <- fa.parallel(TWCorMat, n.obs=0,
fm="ml," n.factors=3, n.iter=500, sim=TRUE)
```

Several specifications must be made when generating the scree plot of eigenvalues and these should mirror those used in the EFA (e.g., correlation matrix, factoring method, number of factors). The number of cases used is specified by n.obs=0, or, in our case, the number of observations can be set to "0" to indicate the data is already formatted as a data matrix. This scree plot used both actual and simulated data generated with parallel analysis, as specified by sim=TRUE. The n.iter=500 syntax specifies the number of simulated analyses to be performed – in this case, we used 30 iterations. After running this syntax, we obtained a scree plot like that shown in Figure 18.2. This plot supports the retention of three factors with our actual data.

The scree plot provides a visual representation of the information produced by the previous list of eigenvalues. A bold horizontal line represents the eigenvalue threshold of 1.0 in the scree plot for easy assessment. The eigenvalues of each factor are represented by an "×" above the factor, with an "×" above the eigenvalue threshold line indicating a retained factor. As can be seen in Figure 18.2, three factors emerged with eigenvalues of 3.81, 1.63, and 1.03, consistent with our previous findings. The remaining factors' eigenvalues all fall below the 1.0 threshold, suggesting that these factors should not be retained in the model (i.e., the construct contains three distinct dimensions). Also shown in Figure 18.2 are the results from a parallel analysis. Originally proposed by Horn (1965), this approach yields eigenvalues from 500 to 1000 simulated data sets of the same size and number of variables. From these random data sets eigenvalues for successive factors are averaged and can be compared to the corresponding eigenvalues from the researcher's substantive-based sample. In Figure 18.2, support is illustrated for three factors – stronger support would have been concluded if the third sample eigenvalue was further above the line from the randomly generated values.

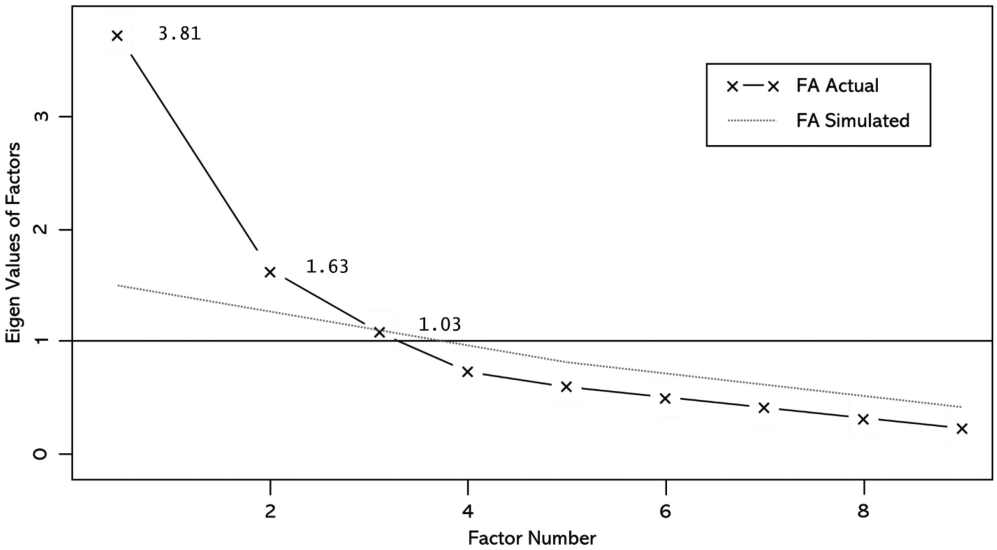


Figure 18.2 Scree plot of eigenvalues for EFA analysis of trustworthiness items

CONFIRMATORY FACTOR ANALYSIS: BACKGROUND

A path diagram for a CFA model for our trustworthiness example can be found in Figure 18.3. Note that this diagram has several similarities with the EFA example, including the use of circles to represent each of the three factors linked to the dimensions of ability, benevolence, and integrity. As with EFA, the factors are linked with the items via factor loadings, and each item is also shown to have unique variance representing random and systematic errors. However, there are some key differences between the EFA and CFA path diagrams given basic differences between the two statistical techniques. First, each of the three factors is shown to have a variance that is estimated (no short arrow with *1.0), in contrast to the assumed factor variance of one that is incorporated in the EFA model. Second, it can be seen that there are many fewer factor loadings because each item is linked only to its intended factor (rather than all three). Finally, unlike EFA where all factor loadings are estimated, in CFA one factor loading for each latent variable is assumed to have a value of 1.0 (shown with *1.0), and the associated item is referred to as the "referent" indicator. The use of a fixed referent factor loading is the default of most CFA software programs and is required for the CFA model to be identified so that a unique set of estimates can be obtained for the following parameters: factor variances, factor covariances, factor loadings, and uniqueness error terms.

In contrast to the EFA, the CFA is implemented using the covariances among the items rather than the correlations, consistent with the underlying statistical theory. Statistical packages for CFA represent a model like the one shown in our figure via a set of measurement equations, one for each item. As an example, the equation for the first Ability item is:

$$[14]\ Ab1 = FL\ (T\text{-}Ab) + 0{*}FL\ (T\text{-}Be) + 0{*}FL\ (T\text{-}In) + u_i$$

Note that as represented in the path diagram, the fact that Ab1 is associated only with the ability factor is indicated by fixing to zero its factor loadings for the benevolent and integrity factors (consistent with these factor loadings not being shown in the CFA path diagram). Comparable equations for the other items, linking each to its respective factor, are also used to represent the model. And, these equations are combined to form a matrix of factor loadings that is used in the CFA estimation process.

The path diagram also indicates that relations are allowed among the three factors (double-headed curved arrows), in the form of factor covariances. These covariances, along with the factor variances that are estimated in CFA (that covariances were correlations in the EFA and were required to be 1.0), are included in a factor variance/covariance matrix. Note that since the factor variances

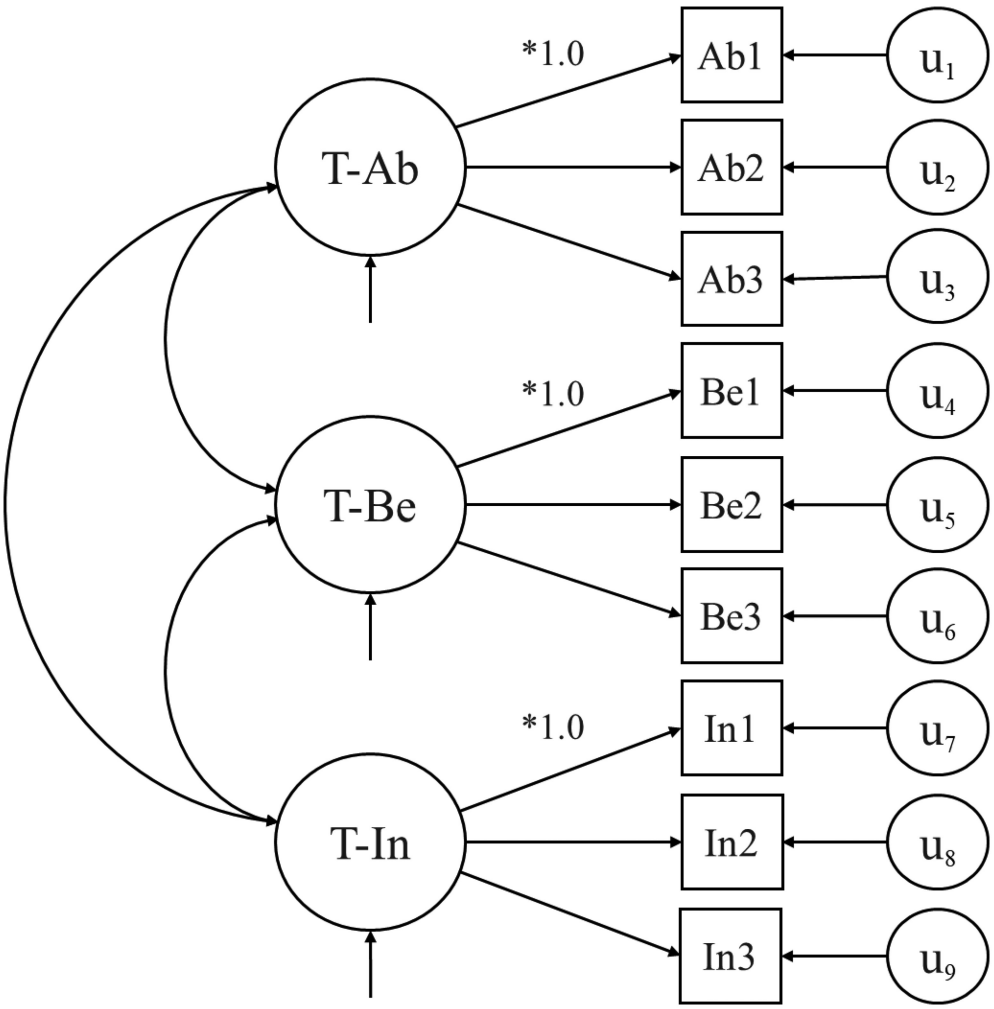


Figure 18.3 Path diagram for CFA model with trustworthiness items

are not standardized, the relations among the factors are covariances and not correlations. Finally, as is typically done in CFA, the uniqueness error terms shown in the path diagram are assumed to be uncorrelated with each other. This is an assumption we made, and unlike EFA which requires this assumption, in CFA some error terms can be allowed to covary when justified theoretically. The variance estimates associated with error terms that are included in the path diagram are represented in the variance/covariance matrix of error terms, but this matrix is a diagonal matrix given our assumption of uncorrelated errors. This assumption is commonly made in applications of CFA but can be relaxed if certain items have common measurement features (e.g., negative wording).

With CFA these parameter estimates are obtained using a process different from EFA. The software programs implement an iterative process, in which estimates described above of factor loadings (FL), factor variance/covariances (FACCOV), and uniqueness covariances (UNICOV) based on the sample item covariance matrix (SAMCOV) are used to compute a predicted covariance matrix (PRECOV) for the items,

$$[15]\ \mathrm{PRECOV}_{9\times9} = \mathrm{FL}_{9\times3} \times \mathrm{FACCOV}_{3\times3} \times \mathrm{FL}^{\mathrm{T}}_{3\times9} + \mathrm{UNICOV}_{9\times9}$$

The differences between corresponding elements of SAMCOV and PRECOV are incorporated into a residual covariance matrix RESIDCOV as:

[16] $RESIDCOV_{9x9} = SAMCOV_{9\times9} - PRECOV_{9\times9}$

The magnitude of these elements of RESIDCOV reveals the quality of the model and its associated parameter estimates. Through an iterative process, the model parameter estimates are revised sequentially using the matrix equations above so that at each step the item residual covariances are made smaller. The process terminates when the estimates cannot be revised so as to make the residual covariances smaller. This final resulting item residual covariance matrix is the basis of determination of model adequacy, with smaller residuals indicating a better model. At this stage standard errors for all the parameter estimates are obtained and can be used for null hypothesis testing. This is in contrast with EFA, where formal statistical hypothesis testing is typically not involved. Unstandardized parameter estimates are initially obtained, since both the item and factor covariances are used, and their standardized versions can also be computed by the software and used to judge their magnitude. As for model adequacy, the item residual covariance matrix forms the basis of the chi-square statistic and various goodness-of-fit indices that have guidelines used to conclude the model is adequate. Currently, the most popular indices and their guideline values include the Comparative Fit Index (CFI > .95), Root Mean Square Error of Approximation (RMSEA < .08), and Standardized Root Mean Residual (SRMR < .10).

CONFIRMATORY FACTOR ANALYSIS: RESULTS WITH TRUSTWORTHINESS

We next implemented a CFA using the same trustworthiness data sample from our prior EFA example. Note that the following syntax will require that the R packages "lavaan," "psych," and "GPArotation" be installed and activated to run. Similar to our approach to EFA, theoryinformed several decisions made while conducting our CFA. We retained our view of trustworthiness as a multidimensional construct consisting of three facets (Frazier et al., 2010) and again used the nine-item trustworthiness measure (three dimensions with three items each) noted in the EFA example from Jones and Shah (2015). Compared to choices with the EFA, in our CFA we followed the common practices of linking each item to only the factor presumed to underly its content, and we also assumed that all measurement errors were uncorrelated with each other.

After checking the data, examining the correlations among items, and creating a data frame to use in the analysis (for a more detailed description, see EFA example), we conducted the CFA. To conduct a CFA of the dimensionality of the trustworthiness items, we used the following:

```
[17] TW <- 'T-Ab =~ Ab1 + Ab2 + Ab3
      T-Be =~ Be1 + Be2 + Be3
      T- In =~ In1 + In2 + In3'
[18] TWCFA <- cfa(TW, data=TWData)
[19] summary(TWCFA, fit.measures=TRUE,
standardized=True)
```

In the above syntax, we specified several key characteristics of our model, beginning with line [17], where we specify how the latent variables are measured. This created an object called "TW" that represents the structure of the measurement model. We specified the items that measure each dimension using the =~ operator. As such, we measured the three dimensions of trustworthiness: T-Ab, T-Be, and T-In using their corresponding clusters of items following our EFA. For example, T-Ab is shown to be linked to items Ab1, Ab2, and Ab3. Note that the use of the + sign does NOT imply that the indicators are combined to a sum – this is simply lavaan notation. With this single line of code, three measurement equations were developed within lavaan (one each for Ab1, Ab2, and Ab3) matching our example equation presented earlier. The syntax from line [18] then implemented the CFA on the trustworthiness measurement model object "TW" using the "TWData" data frame. As discussed earlier, it was necessary for the factor loading of one item to be set to a value of 1.0 for each dimension in order for the model to be identified. The cfa() function conducts several important actions by default, including using the first item of each factor as a referent indicator and fixing its factor loading to a value of 1.0. Also assumed by default with lavaan is that the factors covary, as shown in Figure 18.3, so no code is required to obtain these covariances. Line [19]'s summary() function created an overview of key CFA results and includes a few optional arguments that provided important insight into the adequacy of the model. By including the fit.measures=TRUE argument, additional fit measures were included in the output, such as the model fit statistic (χ^2), degrees of freedom (*df*), comparative fit index (CFI), root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR). The standardized=TRUE argument will produce standardized estimates along with the default unstandardized estimates. The inclusion of these

standardized estimates allows for comparison of corresponding values for factor correlations, factor loadings, and uniqueness values. Our results are presented in Table 18.3.

From the results above, in terms of model adequacy, we followed the practice of focusing on the goodness of fit values rather than the chi-square statistic and its p-value, as the latter is known to be influenced by sample size and deviations from the normality of the items. We reported three important fit statistics: the CFI, RMSEA, and SRMR. Generally, a CFI statistic can be considered acceptable when it exceeds or is close to a value of 0.95 (Bentler, 1990). As our model produced a CFI of 0.939, we judge the model fit to be adequate. Our output also reports a significant

Table 18.3 Results of CFA of trustworthiness items

Model Fit Information						
Model Test User Model:						
Test statistic					135.024	
Degrees of freedom					24	
P-value (Chi-square)					0.000	
User Model versus Baseline Model:						
Comparative Fit Index (CFI)					0.939	
Root Mean Square Error of Approximation:						
RMSEA					0.095	
90 Percent confidence interval - lower					0.080	
90 Percent confidence interval - upper					0.111	
P-value RMSEA <= 0.05					0.000	
Standardized Root Mean Square Residual:						
SRMR					0.065	
Parameter Estimates						
Latent Variables:						
	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
Abil =~						
Ab1	1.000				0.524	0.818
Ab2	1.218	0.069	17.539	0.000	0.638	0.753
Ab3	1.168	0.061	19.173	0.000	0.612	0.862
Bene =~						
Be1	1.000				0.648	0.794
Be2	1.145	0.069	16.698	0.000	0.742	0.884
Be3	0.668	0.053	12.601	0.000	0.433	0.577
Inte =~						
In1	1.000				0.448	0.700
In2	1.159	0.097	11.953	0.000	0.519	0.681
In3	0.963	0.080	11.991	0.000	0.431	0.685
Covariances:						
	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
Abil ~~						
Bene	0.102	0.019	5.392	0.000	0.299	0.299
Inte	0.113	0.015	7.339	0.000	0.481	0.481
Benevolence ~~						
Inte	0.176	0.021	8.376	0.000	0.608	0.608

(*Continued*)

Table 18.3 Results of CFA of trustworthiness items (*Continued*)

Variances:						
	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
Abil	0.274	0.026	10.418	0.000	1.000	1.000
Bene	0.420	0.044	9.645	0.000	1.000	1.000
Inte	0.200	0.026	7.818	0.000	1.000	1.000
.Ab1	0.136	0.013	10.166	0.000	0.136	0.331
.Ab2	0.310	0.025	12.425	0.000	0.310	0.432
.Ab3	0.129	0.016	8.021	0.000	0.129	0.256
.Be1	0.246	0.025	9.782	0.000	0.246	0.369
.Be2	0.154	0.027	5.635	0.000	0.154	0.219
.Be3	0.376	0.026	14.559	0.000	0.376	0.668
.In1	0.208	0.018	11.242	0.000	0.208	0.509
.In2	0.310	0.026	11.724	0.000	0.310	0.536
.In3	0.210	0.018	11.632	0.000	0.210	0.530

RMSEA of 0.095, with a 95% confidence interval of [0.080, 0.111]. While an adequate RMSEA is generally considered to have a magnitude less than 0.08, elements such as small degrees of freedom can influence variance in the RMSEA estimate (see Kenny, Kaniskan, & McCoach, 2015). Finally, our CFA produced an SRMR estimate of 0.065. An SRMR smaller than 0.10 is generally preferred (Shi, Maydeu-Olivares, & Rosseel, 2020), suggesting an acceptable SRMR for our model. It is not uncommon for judgments about model fit to vary across indices used, and we join others who feel that recommended guidelines should not be used too rigidly.

Also important are the results for null hypothesis significance tests for the factor loadings, in the column labeled P(>|z|). The values show that for each of the nine factor loadings, the factor loadings would be considered to be statistically different from zero. In terms of judging the magnitude of all parameter estimates, the column Std. all presents their standardized values. For the factor loading estimates, of the nine values the lowest was .58, two others were .68, and the remaining six standardized factor loadings were all greater than .7. The squared values of these loadings correspond to the communalities from the EFA, and show the percent of variance in each item associated to the single factor it is associated with. As such, these also can be interpreted as the reliability values for each item. As for the factor correlations, they were .30 (T-Ab and T-Be), .48 (T-Ab with T-In), and .61 (T-Be with T-In). Based on all of the information above, we concluded that support was obtained from our CFA for our trustworthiness 3-factor model and that the nine indicators were adequate measures of the three relatively distinct trustworthiness factors.

MODEL COMPARISONS WITH EXPLORATORY AND CONFIRMATORY FACTOR ANALYSIS

Although EFA and CFA employ different mathematical processes, they share in common the principle that a model with a specified number of unobserved factors is proposed to account for correlations or covariances among items included in the analysis. They also share the reality that the decision about any given model and whether it is a good model can be challenging and involves a lot of subjective judgment by the researcher. For this decision, with EFA there are judgments about the number of dimensions present as reflected in eigenvalues (either using a cut-off value of 1.0, a scree plot, or a comparison against randomly generated eigenvalues via parallel analysis). There are judgments about the factor loadings, whether those that are supposed to be large are large enough, and whether others demonstrate simple structure. Judgments also must be made about whether factor correlations are so high that concerns are raised about the discriminant validity of the factors. Alternatively, while the presence of confirmatory in the label of CFA suggests a more rigorous approach to model evaluation, a large amount of judgment is required while using information on the model that is obtained. While a chi-square statistic is available, it is known to be

influenced by sample size and item non-normality and is not used in a formal null hypothesis testing framework. Also, although increased understanding of fit indices such as the CFA, RMSEA, and SRMR has occurred in recent years, the rigid use of cut-off values for model acceptance/rejection has become criticized by many. The main reason is that their values are influenced by features of the model such as the number of items and the number of factors – one set of "gold standard" values can not apply to the wide range of CFA models examined by researchers. Finally, it should also be remembered that despite how favorable the values of any of the EFA/CFA model metrics mentioned above, there is no certainty that the researcher has found the "true" model that actually generated the covariances obtained with the data – there are always alternative models that may provide an equal or better accounting for the shared variance among the indicators.

Given this ambiguity, researchers using EFA and CFA for scale development should extend their decision-making about a proposed model to incorporate information about likely alternative competing models. With EFA this is typically accomplished using models that force extraction of one less and one more factor than the number originally proposed by the researcher. We note that the eigenvalues for these alternative models with the same data will be identical to those originally obtained since they are a property of the correlation matrix and are not influenced by the number of factors extracted (although they are informative about the appropriateness of the number extracted). Instead of eigenvalues, the researcher's focus when examining competing models should be on the pattern of factor loading and the degree to which simple structure is maintained. If the originally proposed number of factors is best, a model with one less factor is examined one or more items will have high values on more than one factor (often referred to as double loading). Also, when one extra factor is extracted the new factor will only have one or two items with high loadings. We examined these alternative models with our trustworthiness data and results are presented in Appendices B and C. These results reveal that with the two-factor model, the clean simple structure of the three-factor model disappeared, and secondary loadings for the T-Be and T-In items were much higher (.20) than with the three-factor model for which no secondary loadings were greater than .08. Alternatively, with the four-factor model the added factor had only a single item with a high factor loading values. Thus, results for the alternative two- and four-factor results are not as favorable as those for the proposed three-factor model, adding to confidence in the favorable conclusion of the three-factor model.

With CFA, model comparisons are equally as important, and in the present context, the alternative model that is most typically examined is one that proposes a single factor instead of the proposed number of factors (in our case three). This comparison is often framed as an overall test of the discriminant validity among the proposed multiple factors since the single-factor model can be understood to impose restrictions that the correlations among the multiple factors are all equal to 1.0. In our case, the three correlations among the T-Ab, T-Be, and T-In factors are no longer estimated and are assumed to be one, and this results in the new more restrictive model having a higher chi-square value and more degrees of freedom. And since these two models can be seen as nested, they can be compared using a chi-square difference test. The CFA results for this one-factor model are provided in Appendix 18.D, and it can be seen that for the one-factor model, the chi-square increased to 773.00 and the degrees of freedom increased by three to 27 since the three previously estimated factor loadings were constrained to 1.0. The difference in the two values for the three-factor and one-factor models of 637.98 exceeds the critical .05 chi-square value for three degrees of freedom, and the null hypothesis that the three-factor correlations are 1.0 is rejected and the original three-factor model is retained. However, it should be noted that even when the hypothesis that the factor correlations are 1.0 is rejected, the estimated factor correlations can still be so high that the discriminant validity of one or more of the factors is challenged. This is not the case with our three trustworthiness factors, which were only moderately correlated.

ADVANCED ISSUES WITH EXPLORATORY AND CONFIRMATORY FACTOR ANALYSIS

The preceding sections on EFA and CFA have introduced the conceptual background of these types of models and their statistical underpinnings and demonstrated with a simple example their use in scale development and evaluation. Those seeking a deeper understanding of EFA/CFA as they plan to use the techniques in their own scale development work need to be well informed on several issues that may emerge as they proceed. Space limitations preclude in-depth discussion of these issues, so instead, we will share resources we have found helpful in our own research and teaching. Hair, Black, Babin, and Anderson (2019)

have a very complete and understandable chapter on EFA in their book *Multivariate Data Analysis* that covers all aspects of its use.

For example, they discuss the use of Bartlett's test of sphericity and the Kaiser-Meyer-Olkin approach to determining if sufficient correlations exist to justify the use of EFA. Also, while in our trustworthiness example, we used maximum likelihood as the estimation method to match our CFA approach, there are several extraction techniques that others may want to consider, and Hair et al. (2019) provide a good comparison of these techniques. They also give good coverage of the use of the scree plot and parallel analyses that can facilitate understanding of their role in determining the number of factors to be retained. While we suggested with our example that assuming factors are correlated will typically be theoretically appropriate, Hair et al. (2019) also discuss the choice of oblique (correlated) vs. orthogonal (uncorrelated) rotation. Finally, interpretation of factor loadings is covered, and they suggest that minimum values are .30-.40, practically significant values are in the .50 range, and well-defined structures have loadings around .70. They further guide that as sample size and the number of variables increases, smaller loadings become more acceptable. They also discuss the value of comparing squared loadings to compare items from within an analysis. Finally, Hair et al. discuss the role of factor loadings in the creation of scales and link all the above topics together with an integrative example.

Those seeking a more rigorous math presentation of EFA may favor the Tabachnick and Fidell (2019) chapter from their *Applied Multivariate Statistics* book. They note that factor analysis will be more successful with some item correlations greater than .30, and for sample size, they report that with data where items have a high communality with many indicators, samples of 100–200 may be adequate, with 300 or more being required when there are a small number of factors and 3–4 indicators per factor. Example topics are addressed with greater use of mathematical explication including details on factor extraction and rotation and a comparison of available techniques for both steps. For oblique rotations, they note that pattern matrices, in which regression-like coefficients linking items to factor will typically be easier to interpret, while adding that loadings of .45 or higher are desirable. Also considered are practical issues related to normality, outliers, and multicollinearity. Like Hair et al., they include a comprehensive example, with details on several software packages used with the example.

The books above give similar detailed treatment of CFA that is a good starting point for a more detailed understanding of its use in survey development. Hair et al. (2019) have a separate chapter on CFA that covers the stages of defining constructs, developing the measurement model, study design, and assessing model validity. They include discussion of the use of average variance extracted (AVE) and composite reliability diagnotics for judging quality of latent variable measurement, with a complete example that illustrates use of all steps of a CFA. Alternatively, even though Tabachnick and Fidell (2019) include CFA as part of a broader chapter on structural equation modeling, they go into more detail about various aspects of CFA. For example, Tabachnick and Fidell (2019) provide a deeper treatment of the matrix processes underlying parameter estimation with CFA and a more complete treatment of indices used in model fit and model evaluation. They also provide a detailed comparison of available software programs for CFA.

Of special relevance to researchers using CFA in theoretically driven or applied research that develops scales are two new articles in *Organizational Research Methods*. Nye (2022) has discussed issues he views as essential to CFA applications, providing a nice summary of technical literature outside the normal scope of organizational researchers. Nye (2022) first reviews CFA parameter estimation, noting the wide use of maximum likelihood and its assumptions, including that variables in the model are normally distributed. Nye (2022) also describes alternative estimators, noting that diagonally weighted least squares is generally preferred. A second issue of focus for Nye is power analysis to guide sample size determination. Three approaches are summarized and their implementation described, with discussion of the match of the approach to the goals of the study. A third topic covered by Nye (2022) is approaches to missing data. After contrasting listwise and pairwise approaches, he notes the importance of whether data is missing randomly or not. He then contrasts techniques of full information likelihood (FIML) and multiple imputation (MI), noting that both can be effective but that FIML is popular in many packages used for CFA. Finally, Nye (2022) recommends that for model fit, chi-square and traditional fit values should be reported, all measures should be considered, and additional characteristics of the model should be used to give a more complete picture of model adequacy.

The second paper (Ronko & Cho, 2022) reviews issues related to discriminant validity, which as noted earlier involves determining whether a factor underlying one set of items is distinct from a factor associated with a second set. In our section on model comparison, we illustrated a frequently used approach of testing the hypothesis that a

factor correlation is 1.0, and noted that even if the researcher rejects this hypothesis the factor correlation may still be so large that concerns about the distinctiveness of the two factors remain. Ronko and Cho (2022) begin by reviewing how the concept of discriminant validity is understood in at least two different ways, and as a result, empirical procedures based on the model comparison just described vary widely in us. They then describe a synthesized meaning of discriminant validity and report on two techniques, one that is based on the confidence intervals around estimated factor correlations and the other that focused on the model comparison. After conducting a simulation study of multiple approaches, they then present a workflow diagram that can guide researchers to the models and results most useful for discriminant validity assessment. They also include a Supplement with a tutorial on how to implement the techniques in a variety of CFA software.

COMMENTS ON COMBINED USE OF CORRELATIONS AND EFA/CFA

We restate the typical objective with survey research is to develop a set of items to form a scale. To overcome the challenges of eyeballing patterns of correlations among a large number of items, EFA and CFA allow the researcher to propose a theory-based expected pattern and then examine evidence that the pattern is obtained with the data. In our example, the pattern could be seen in the actual individual correlations (given a small number of factors and items for each), and this resulted in evidence confirming that three dimensions were present and the items proposed for each dimension were measuring the intended construct. Evidence of the former included eigenvalues (pattern when plotted and compared to random data in parallel analysis), simple structure in the factor loadings, and factor correlations that are not so high as to raise discriminant validity concerns.

A pattern of item correlations that does not meet the structure below is what drives problems that emerge in the factor analysis. For example, with EFA a lack of strong within-dimension correlations contributes to low factor loadings for the items on their factor and lowers the associated eigenvalue and variance accounted for by the factor. For the corresponding CFA, factor loadings will also be low and this will lower values for the Average Variance Extracted and Composite Reliability metrics that reflect latent variable measurement. It may also increase the magnitude and significance of the standardized residual covariances for the within-dimension set, as the lack of share variance due to the factor may leave more covariance unexplained. Alternatively, when the between-dimension item relations increase, this makes it more difficult for EFA-obtained eigenvalues to support additional factors and the relative amount of explained variance due to the factors can decrease. More apparent will be a lack of simple structure in the factor loading pattern matrix and/or increasing factor correlations. In the CFA for this data, goodness of fit will be lowered because the moderate- to high-value secondary loadings are specified to be zero, resulting in relatively less of the covariances among between set items being accounted for. And, large and significant between-factor item residual covariances are likely to occur. In short, failure to obtain desired EFA/CFA results begins with problems with the correlations among indicators, and review of these correlations can help researchers anticipate, prevent, and understand why the desired results were not obtained.

One special issue of importance that has not been addressed relates to whether it is appropriate to conduct EFA and CFA on the same data set during the scale development process. Discussion and comparison of the two techniques have occurred over the past 25 years (e.g., Hurley, Scandura, Schreisheim, Brannick, Seers, Vandenberg, & Williams, 1997; Williams, 1995). In considering this issue, it is important to remember that although EFA and CFA share certain features, including the assumption that scores on items are a function of something unobserved or latent in the form of a factor, they are different statistical techniques. The underlying computations involved are different, as are the resulting parameter estimates and model diagnostic information. More importantly, both EFA and CFA statistical models can be used in either an exploratory or confirmatory approach. We have emphasized the use of EFA in a confirmatory way, in which the research starts out with an a-priori specification of the number of factors (in our case three) and then looks for information confirming that specification (using eigenvalue information, pattern of factor loadings, and size of factor correlations). We also note that CFA can be used in an exploratory fashion, and this approach is taken often by organizational researchers. An example of this would be when a researcher examines a model with a set of items and a specified number of factors but then learns their model does not fit well and revises it by dropping items and/or respecifying the number of factors. The real problem occurs when the researcher then suggests that they have "confirmed" their new model, after revising it based on analyses with the

original data. This process runs the risk of capitalizing on chance.

Clearly, the preference when developing scales is for a researcher to have multiple samples so that they can obtain results from one sample, then test any revised model with additional samples. This increases the likelihood that the conclusions confirmed from the revised model will be replicated in future samples. However, we believe that in such a process there is a role for both EFA and CFA at both stages, in the original and replication samples. Both techniques give different and complementary types of information about the number of dimensions, the strength of the link of items with primary (intended) and secondary (unintended) factors, and the relations among factors. Examining results from both techniques for each of these topics of interest overcomes the limitations of each. For example, using eigenvalues to judge the number of dimensions is supported by examining the fit of the corresponding CFA model with its specified number of dimensions. Similarly, secondary loadings for items on factors other than the ones they are intended to measure can be directly examined with an EFA, with the researcher hoping that they are small in magnitude. Alternatively, with CFA evidence that the link between items and secondary factors are small is obtained by the good fit of a model that forces them to zero. Most importantly, the key point is that any time a model is revised based on the use of a specific sample, that same sample can not be used to confirm any changes to the original model, because the revisions will be guaranteed to fit well. This is true with either EFA or CFA. Independent confirmation of the model with a second sample, using both EFA and CFA, is needed before it can be viewed with confidence.

Finally, we encourage researchers using EFA/CFA models on a regular basis in scale development to regularly scan quantitative methods journals for the latest developments that may impact their analyses. For example, Auerswald and Moshagen (2019) have investigated how the determination of the number of factors can be impacted by extraction method, including several methods not covered in this review. Based on their results, they recommend a combined approach as best for choosing the number of factors to retain. Alternatively, Goretzko (2022) has studied the impact of missing data on factor retention, concluding that the missing data choice should be based on features of the data and which factor retention rule will be used. Other researchers have investigated using CFA fit measures with EFA models. For example, Finch (2020) compared CFA fit measures to traditional EFA criteria and found the former to have potential with categorical indicators or when loadings are low with metric indicators. In the same topic area, Montoya and Edwards (2021) found the usefulness of fit measures to depend on the degree of model misspecification. A final area worth mentioning is the development of new factor-based models. In a review of factor analysis in psychological assessment, Sellborn and Tellegen (2019) identify new exploratory structural equation modeling (ESEM) techniques and bi-factor models as having potential for increasing understanding of model structure, while Steenkamp and Maydeo-Alvares (2022) introduce "unrestricted" factor analysis, similar to ESEM, as a combination of EFA and CFA with potential for investigating data structures.

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APPENDIX 18.A EXAMPLE CORRELOGRAM

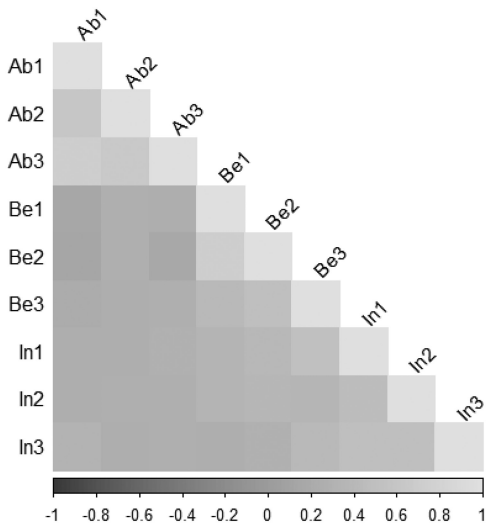


Figure 18.A1 Correlogram of nine-item trustworthiness measure

APPENDIX 18.B EFA RESULTS OF A TWO-FACTOR MODEL OF TRUSTWORTHINESS

Items	*Factor 1*	*Factor 2*	*Communality* (h^2)	*Uniqueness* ($1-h^2$)
Factor1	**(1.00)**			
1. Ab1	0.85	–0.06	0.69	0.31
2. Ab2	0.71	0.11	0.56	0.44
3. Ab3	0.85	0.00	0.73	0.27
Factor 2	**(0.32)**	**(1.00)**		
4. Be1	0.00	0.78	0.60	0.40
5. Be2	–0.08	0.89	0.75	0.25
6. Be3	0.13	0.56	0.38	0.62
7. In1	0.21	0.45	0.31	0.69
8. In2	0.21	0.41	0.27	0.73
9. In3	0.26	0.38	0.25	0.75
Eigenvalue	3.81	1.63		
Proportion of variance	0.26	0.25	0.51	0.49

Figure 18.B1 EFA results of two-factormodel of trustworthiness

APPENDIX 18.C EFA RESULTS OF A FOUR-FACTOR MODEL OF TRUSTWORTHINESS

Items	*Factor 1*	*Factor 2*	*Factor 3*	*Factor 4*	*Communality* (h^2)	*Uniqueness* ($1-h^2$)
Factor 1	**(1.00)**					
1. Ab1	0.82	–0.08	–0.02	0.07	0.68	0.32
2. Ab2	0.74	0.14	–0.01	–0.04	0.58	0.42
3. Ab3	0.86	–0.01	0.04	–0.02	0.75	0.25
Factor 2	**(0.26)**	**(1.00)**				
4. Be1	0.07	0.79	–0.02	–0.01	0.64	0.36
5. Be2	–0.04	0.88	0.04	0.03	0.82	0.18
Factor 3	**(0.27)**	**(0.47)**	**(1.00)**			
6. Be3	0.01	0.08	0.89	0.01	0.91	0.09
Factor 4	**(0.44)**	**(0.51)**	**(0.54)**	**(1.00)**		
7. In1	0.02	0.05	0.20	0.52	0.48	0.52
8. In2	0.01	0.12	–0.08	0.65	0.46	0.54
9. In3	0.01	–0.06	0.02	0.75	0.55	0.45

Figure 18.C1 EFA results of four-factor model of trustworthiness

APPENDIX 18.D CFA RESULTS FOR ONE-FACTOR TRUSTWORTHINESS ITEMS

Model Test User Model:	
Test statistic	773.001
Degrees of freedom	27
P-value (Chi-square)	0.000
User Model versus Baseline Model:	
Comparative Fit Index (CFI)	0.591
Tucker-Lewis Index (TLI)	0.454
Root Mean Square Error of Approximation:	
RMSEA	0.233
90 Percent confidence interval - lower	0.219
90 Percent confidence interval - upper	0.247
P-value RMSEA <= 0.05	0.000
Standardized Root Mean Square Residual:	
SRMR	0.131
Parameter Estimates:	

(*Continued*)

(*Continued*)

Latent Variables:						
	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
TW =~						
Ab1	0.341	0.029	11.868	0.000	0.341	0.532
Ab2	0.482	0.038	12.843	0.000	0.482	0.569
Ab3	0.412	0.031	13.145	0.000	0.412	0.580
Be1	0.475	0.036	13.203	0.000	0.475	0.582
Be2	0.512	0.037	13.988	0.000	0.512	0.610
Be3	0.472	0.033	14.513	0.000	0.472	0.629
In1	0.410	0.028	14.906	0.000	0.410	0.642
In2	0.449	0.033	13.406	0.000	0.449	0.589
In3	0.373	0.028	13.494	0.000	0.373	0.593
Variances:						
	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
TW	1.000			1.000	1.000	
.Ab1	0.294	0.020	14.678	0.000	0.294	0.717
.Ab2	0.485	0.034	14.405	0.000	0.485	0.677
.Ab3	0.334	0.023	14.311	0.000	0.334	0.664
.Be1	0.440	0.031	14.293	0.000	0.440	0.661
.Be2	0.442	0.032	14.026	0.000	0.442	0.628
.Be3	0.341	0.025	13.827	0.000	0.341	0.605
.In1	0.240	0.018	13.667	0.000	0.240	0.588
.In2	0.378	0.027	14.227	0.000	0.378	0.653
.In3	0.257	0.018	14.198	0.000	0.257	0.649